

Matt Parrett

Software engineer with 15 years of experience, focused on building ML platforms for the last decade at Amazon Alexa, Mercari, and Sprout Social. I make ML systems fast, cheap, and safe in production — most recently cutting transformer inference costs 35% while 5x'ing throughput.

✉ matt.parrett@gmail.com
☎ +1 520-271-0930
in in/mattparrett
📍 Bothell, WA

PROFESSIONAL EXPERIENCE

Sprout Social, Engagement & AI/ML Platform | January 2023 - Present Senior ML Platform Engineer

Developed ML platform, services, and infrastructure automation for the Sprout Social app.

- Scaled ML infrastructure from single service to 10+ inference services through reusable component development, enabling critical product features and supporting company-wide AI initiatives.
- Reduced transformer inference GPU Utilization and AWS spend by 35% while growing from 2K to 10K+ messages/second via automated testing, GPU monitoring, batch optimization, and capacity planning. Led PII Service productionization for AI safety compliance.
- Automated complex operational workflows, transforming multi-week manual Elasticsearch rebuilds into automated processes completed in days.

Mercari, US Machine Learning Platform | November 2021 - October 2022 Senior ML Platform Engineer

Developed ML platform and infrastructure for the Mercari shopping app. Provided cross-functional engineering support for science/ML teams.

- Enabled real-time recommendation infrastructure by integrating Vertex AI Matching Engine's pre-GA streaming feature, reducing data update latency from hours to minutes for 10+ million user vectors.
- Delivered measurable business impact through model optimization, achieving +5% absolute accuracy in shipping predictions and >10% GMV improvement in recommendation services through A/B testing and production deployment.
- Decoupled and refactored a Prophet-based service, increasing anomaly detection frequency for CRM data from 1 hour to ~1 minute.

Amazon, Alexa Secure AI Foundations | August 2017 - July 2021 Software Dev. Engineer II

Designed and developed services and SDKs for a distributed ML platform.

- Developed a metadata service, web portal, and AWS account creation API enabling secure role-based data access and self-service onboarding for 1,500+ science users, reducing onboarding time from 2 weeks to under 1 hour.
- Enabled secure collaboration at scale through Python SDK development for ML model repository, facilitating secure sharing of thousands of job artifacts across science teams and integrated metrics ingestion across ML frameworks.
- Mentored junior engineers and conducted technical interviews during rapid growth in Alexa (4x headcount).

CORE SKILLS

ML & Inference

Transformer Inference
Optimization, MLOps,
Machine Learning, Data
Engineering

Languages

Python, Java, TypeScript

Cloud & Infra

AWS, GCP, Linux/Shell, SQL

Engineering Practices

System Design,
Performance Optimization,
CI/CD, Monitoring,
Mentoring

Frontend

React, JavaScript

ABOUT

BJJ practitioner; tinker with binary neural networks and genealogy on the side.

PROFESSIONAL EXPERIENCE

Broken Bulb Game Studios | November 2009 - October 2016

Lead Engineer, Technical CTO

- Guided software engineering and architecture including back-end and front-end teams across 10 mobile/social game launches with 20+ million monthly active users.
- Designed and developed the infrastructure, front-end game engine, world editor, and player-versus-player game server for the game Miscrits handling 1,000's of concurrent real time games.
- Saved 70% on infrastructure costs by migrating game services to cloud infrastructure with minimal downtime (<1 hour).

EDUCATION AND COURSEWORK

- Arizona State University – Mathematics coursework supporting ML (2019–2020)
- University of Arizona (1999–2000)
- Selected coursework: Machine Learning (Stanford/Coursera), Neural Networks & Deep Learning (deeplearning.ai), Machine Learning for Trading (Google Cloud/NYIF)